

Collaborative Filtering - based Recommendation System.

pseudo code

	user 1	user 2	...	user n
book 1	5	2		
book 2	3	5		
book 3				
...				
book n				

users > 250
books > 50

* Taking already those user which rates books more than > 250

Here, now we need experienced users

We are values rating of that users which are valid and which are experience.

→ We consider that books only that atleast 50 user read and rate it.

If we group by book-title so it means how many users rated each book

Date: / / Page no:

ratings - with - name - group by ('User-ID'). count()
['Book-Rating'] > 2000

→ now do this

group by ('User-ID')

group by user, books

group all rows having same USER-ID
Find all the

count() → count the no. of non empty values
in each column for every group
→ count books for all user.

['Book-Rating'] → show me only.
Book rating data

→ Find all the books group by user > 2000.

→ Find all the books - rating group by user.
200+
→ I will save user total & how many book rated
for & /

$x = \text{rating_with_name.groupby('User-ID')}$
 $\text{count()['Book-Rating'] > 200}$

people_vote_books = x

Out	User ID		} To make True
	2	F	
	8	F	
	9	F	

people_vote_books = x[x].index

filtered_rating = rating_with_name
 $[\text{rating_with_name['User-ID'].isin}$
 $(\text{users_vote_books})]$

isin() Is this value present in users_vote_books.

Ex =	User ID		User vote to
	101	X	102
	102	✓	109
	103	X	

102 in dataframe

$y = \text{filtered_rating} = \text{filtered_rating} \cdot \text{groupby}$
 $(\text{'Book-Title'}) \cdot \text{count}() [\text{'Book-Rating'}]$
 50

$[\text{'Book-Rating'}] >= 10$

Put all the same books together
 count how many rating each book have
 and

count how many rating each book have and
 mark true for books with at least 50

ratings

$\text{filtered_rating} [\text{filtered_rating} [\text{'Book-Title'}] \text{ is in (famous-books)}]$

False

$y[y]$
 True

pt.fillna(0, inplace=True).

Date: / / Page no:

pt = binal_ratings.pivot_table(index='Book-Title', columns='User-ID', values='Book-Rating')

* pivot-table convert the data into table

* columns = 'User-ID'
Create one column for every user

* values = 'Book-Rating'
Fill each cell with the rating given by the user for that book.